Unsupervised Clustering of Navigation Environments for Adaptive Policy Switching

Final Project Report

Session Spring 2026

Course: Machine Learning For Robotics (MLR)

BSc. in Computer Science (Robotics)

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Declaration: The work presented in this report was completed end-to-end using an automated machine learning pipeline and represents the findings of the group using Moving AI benchmarks.

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Abstract

This project studies navigation environments using structural grid features, unsupervised clustering, supervised classification, regression on path length proxies, and tabular reinforcement learning (RL) for adaptive policy switching. We parse 260 MovingAI grid maps across four environment families (maze, room, random, street), compute 17 geometry-based features, and generate labels for map type, difficulty, and obstacle density. We train and evaluate clustering (K-Means), classification (KNN, SVM, decision trees, bagging, boosting), and regression models (BFS-based average path length, RL-derived average steps to goal). A cluster-specific Q-learning policy is compared with a universal baseline for zero-shot transfer. Results show strong separability for map types using geometric features, mixed clustering structure emphasising navigability rather than map names, and limited generalisation for tabular RL policies under distribution shift. We provide comprehensive artifacts: metrics, figures, trained models, and reports in the outputs directory.

1 Introduction

Navigation in grid-based environments demands generalisation across diverse geometric structures. This project explores whether unsupervised clustering of environment geometry can support adaptive policy selection and whether structural features can reliably predict environment class, difficulty, and navigation cost. We integrate classical ML, graph search, and tabular RL into a single reproducible pipeline covering data extraction, feature engineering, label generation, clustering, classification, regression, RL training, and adaptive policy evaluation.

2 Dataset

We use the **MovingAI Grid Benchmark** datasets (`https://www.movingai.com/benchmarks/grids.html`), which comprise 260 grid-based maps spanning four distinct families:

- **Maze** (60 maps): Generated as `maze512-{W}-{ID}`, where the corridor width $W \in \{1, 2, 4, 8, 16, 32\}$ controls navigational complexity.
- **Room** (40 maps): Structured indoor layouts of the form `{N}room_{ID}`, where $N \in \{8, 16, 32, 64\}$ denotes the number of rooms.
- **Random** (70 maps): Stochastic obstacle fields defined as `random512-{D}-{ID}`, with obstacle density $D \in \{10, 15, 20, 25, 30, 35, 40\}$.
- **Street** (90 maps): Real-world city maps (e.g., Berlin, Boston, Shanghai) at multiple resolutions (256, 512, 1024), denoted as `city_{variant}_{size}`.

All labels and regression targets are derived programmatically from map structure or filename metadata. No ground-truth labels are externally provided.

Each map is represented as a 2D occupancy grid, where free cells are encoded as 0 and obstacles as 1.

3 Feature Engineering and Labels (Phases 1A–1B)

For each map, we compute a set of 17 structural features capturing spatial complexity, connectivity, and geometric properties. These include:

- Obstacle density and free-space ratio
- Connected component statistics (count, average size, largest component)
- Entropy of obstacle distribution
- Horizontal and vertical symmetry
- Border obstacle ratio
- Grid aspect ratio

A composite difficulty score is defined as:

$$d = 0.40 \cdot \rho + 0.30 \cdot \frac{1}{\bar{C} + 1} + 0.30 \cdot \frac{1}{C_{\max} + 1} \tag{1}$$

where ρ denotes obstacle density, $\bar{C}$ is the average connected component size, and $C_{\max}$ is the size of the largest free-space component.
Difficulty labels (*easy*, *medium*, *hard*) are assigned via percentile-based binning of the difficulty score. Obstacle density is categorized into *low*, *medium*, and *high* using fixed thresholds.

Additional metadata, such as corridor width, room count, obstacle percentage, city identifier, and resolution, is extracted from filenames where available.

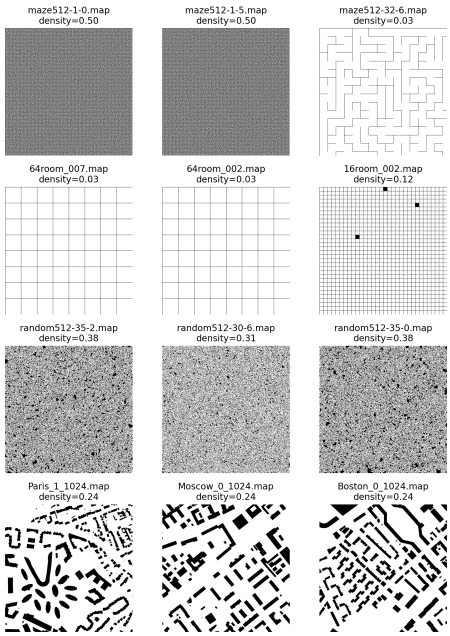

Figure 1: Sample maps categorized by type and obstacle density.

4 Phase 0–2: Setup, Parsing, and Preprocessing

We extracted all archives, verified map counts (maze = 60, room = 40, random = 70, street = 90), and generated the feature matrix (260×17) with no missing values. Labels were validated using known densities and corridor widths. Preprocessing included standard scaling and PCA (10 components explaining 99.14% variance). Artifacts were saved in `outputs` and `data/processed`.

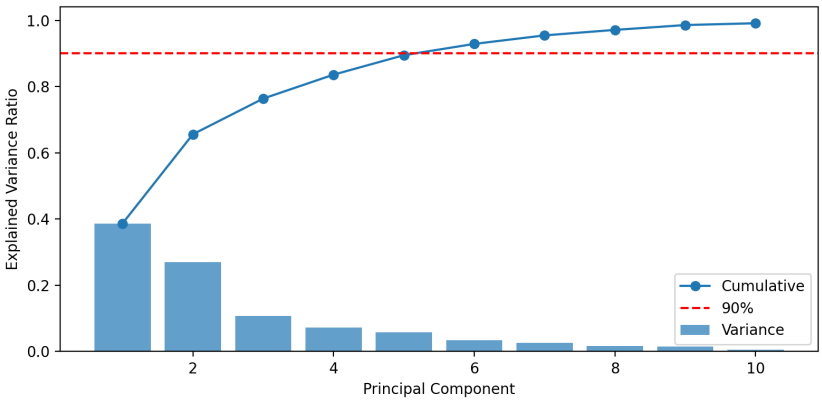

Figure 2: PCA cumulative explained variance.

5 Phase 3: Unsupervised Clustering

We evaluated K-Means with $k = 2 \ldots 10$ using inertia, silhouette, and Davies–Bouldin. The silhouette optimum was $k = 2$ (0.527), but $k = 4$ was selected for interpretability and downstream RL alignment. With $k = 4$, cluster sizes were $\{0{:}100,\ 1{:}30,\ 2{:}80,\ 3{:}50\}$. Cross-tabs showed that clustering captures navigability structure rather than map-type labels.

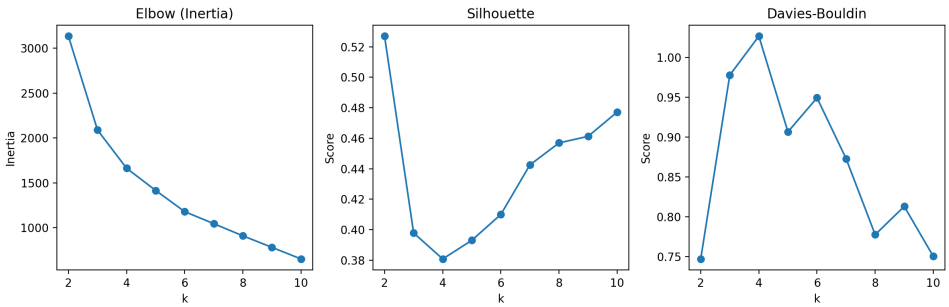

Figure 3: K selection metrics across k values.

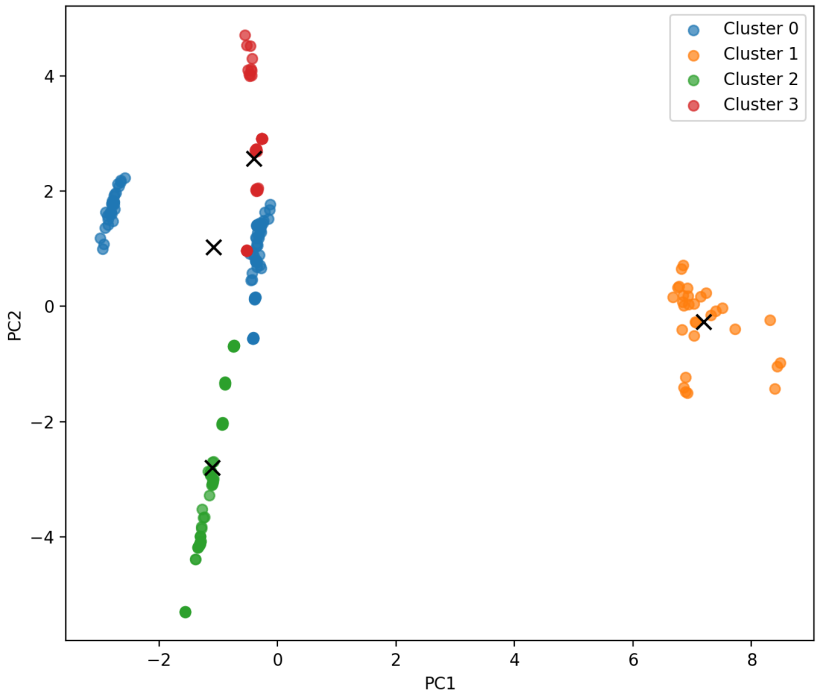

Figure 4: 2D PCA visualisation of $k = 4$ clusters.

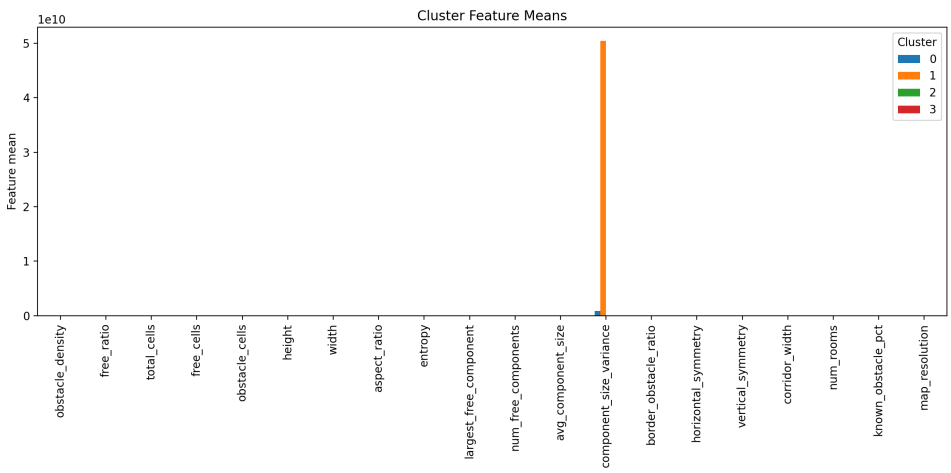

Figure 5: Cluster feature profiles (normalised).

6 Phase 4: Supervised Classification

We trained KNN, SVM, decision tree, bagging, and AdaBoost classifiers on geometry-only features. Phase 4 results show near-perfect performance for map type and density categories, with strong difficulty classification. Leakage checks passed (no duplicate maps, no train/test overlap, permutation sanity near chance).

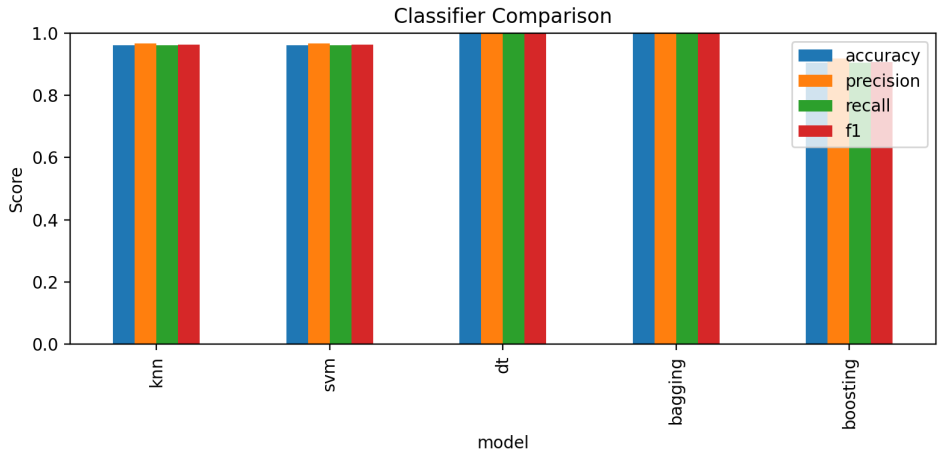

Figure 6: Classifier comparison across targets.

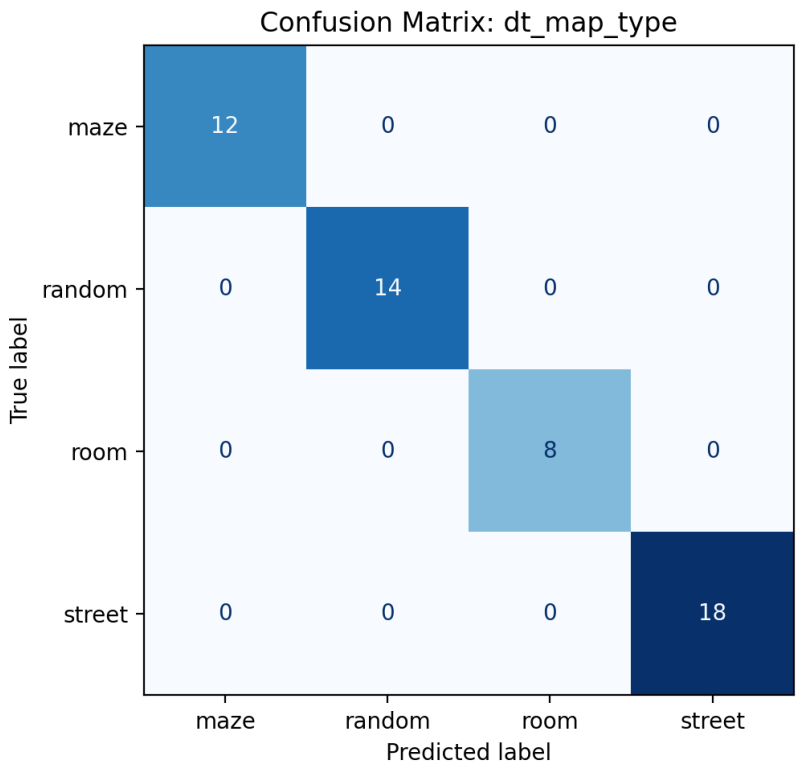

Figure 7: Decision tree confusion matrix for map type.

7 Phase 5: Regression on BFS Path Length

Average shortest path length was approximated using BFS over downsampled grids (50 random pairs per map). Coverage was 219/260 maps; missing targets clustered in extremely dense or disconnected maps. Linear regression provided a strong baseline; polynomial regression was unstable due to overfitting.

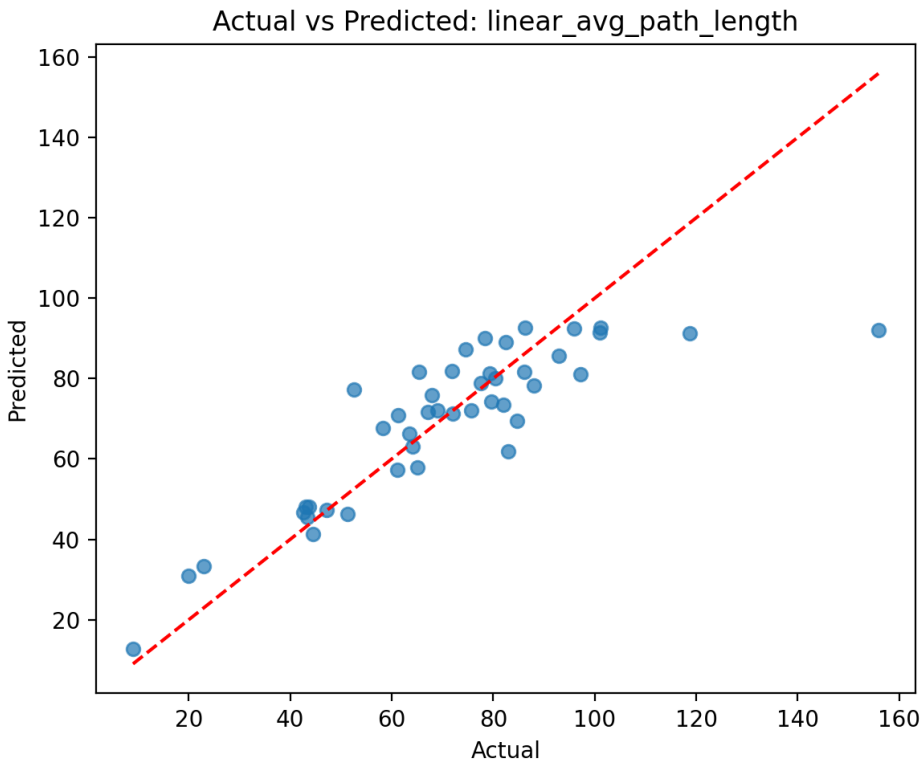

Figure 8: Linear regression: actual vs. predicted average path length.

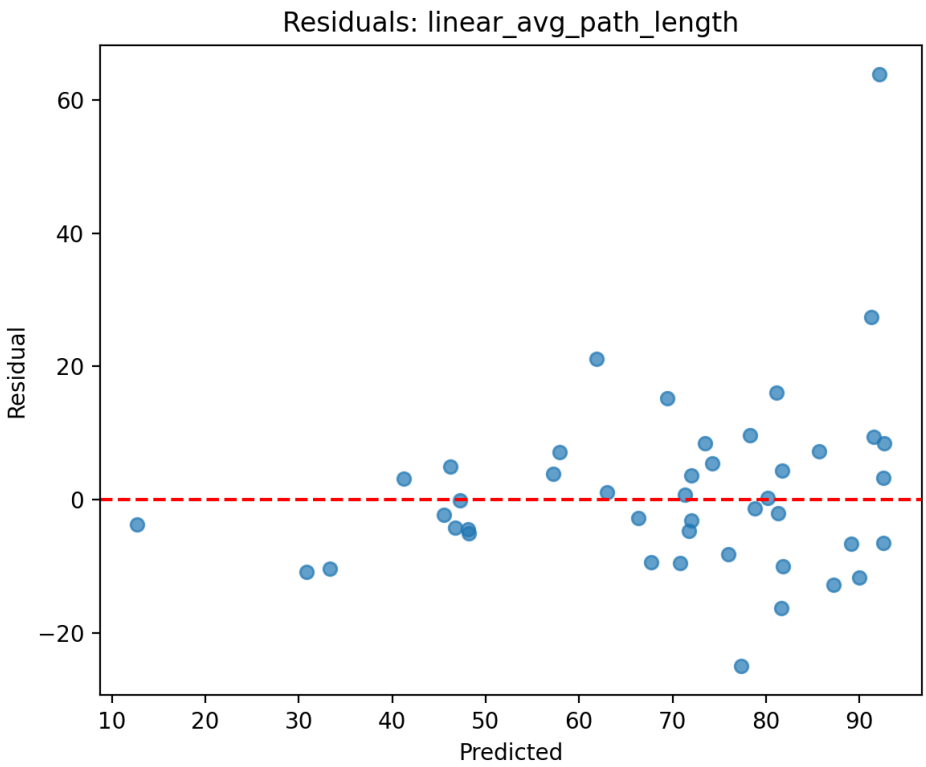

Figure 9: Linear regression residuals for average path length.

8 Phase 6: Tabular RL for Steps-to-Goal

Cluster-specific Q-learning agents were trained on downsampled grids with reward shaping and a curriculum schedule. While training showed intermittent progress, greedy evaluation on unbiased random start–goal pairs failed across clusters, indicating poor generalisation and curriculum overfitting. RL-derived targets were therefore treated as a weak auxiliary signal rather than a reliable planner.

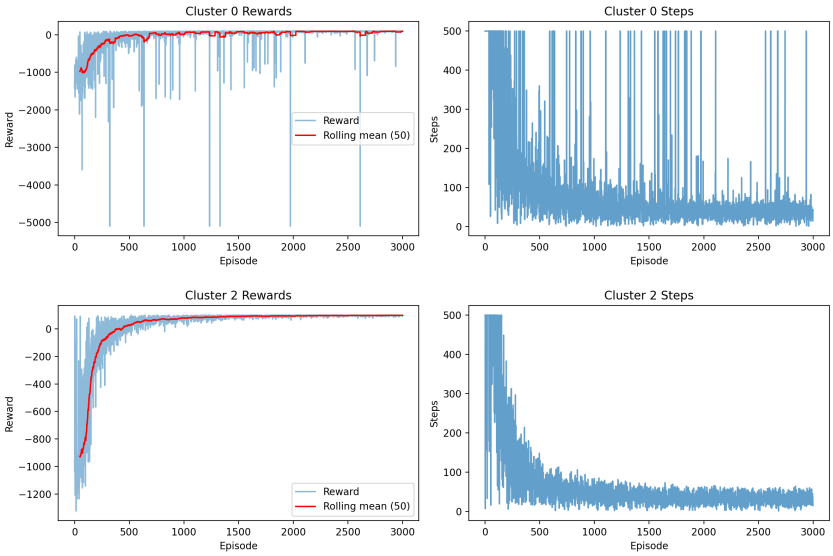

Figure 10: Training curves for two representative clusters.

9 Phase 7: Adaptive Policy Switching

We compared cluster-specialised policies with a universal baseline trained on the PCA-centroid map (`Denver_1_512.map`). A strict hold-out test set excluded training maps and corridor-width-1 mazes. The adaptive policy showed a clear advantage on structurally similar maze maps (26% vs. 6% success on `maze512-8-5`), while both policies failed on disconnected rooms and high-entropy random maps.

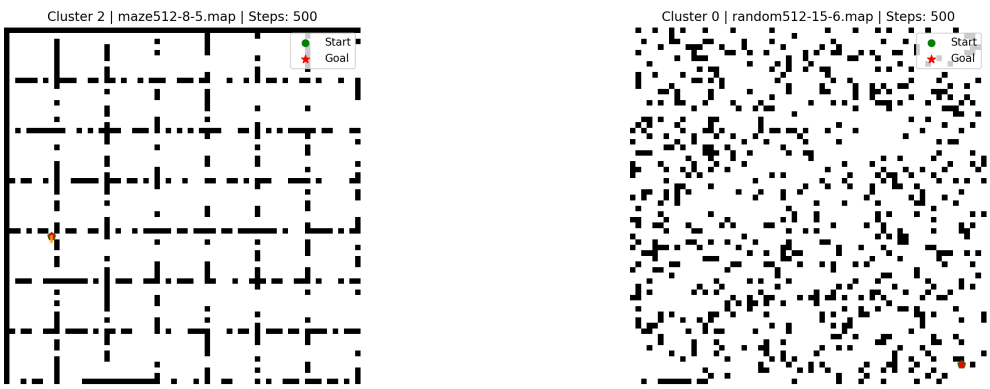

Figure 11: Policy paths on representative maps (adaptive vs. baseline behaviour).

10 Phase 8: Final Evaluation and Complexity

Final consolidated metrics (Phase 8) summarise classification, regression, clustering quality, and RL profiling. Classification for map type remains strong, while difficulty and density are moderate under strict evaluation. Regression on BFS-derived path length remains usable ($R^2 \approx 0.72$), while RL-derived targets are weak ($R^2 \approx 0.52$). Clustering validation yields silhouette 0.381 for $k = 4$. End-to-end pipeline runtime was approximately 87 seconds for all phases.

10.1 Classification (Phase 8)

Model	Accuracy	Precision	Recall	F1
KNN (map type)	0.962	0.966	0.962	0.961
SVM (map type)	0.962	0.966	0.962	0.961
Decision Tree (map type)	1.000	1.000	1.000	1.000

Table 1: Selected map type classification results (see `outputs/reports` for full tables).

10.2 Regression and Clustering (Phase 8)

Target	RMSE	MAE	R^2
Avg path length (linear)	15.924	12.103	0.717
Avg steps to goal (linear)	96.030	67.224	0.518

Table 2: Regression metrics for BFS and RL targets.

Silhouette	Davies–Bouldin	Calinski–Harabasz
0.381	1.027	165.888

Table 3: Clustering validation for $k = 4$.

11 Discussion

The pipeline demonstrates that grid-derived structural features are highly predictive of map type and density, and that clustering captures meaningful navigability groups even when not aligned with map labels. Regression on BFS-derived path lengths produces a usable proxy for navigation cost; however, tabular RL struggles to generalise under distribution shift despite curriculum training. Adaptive policy switching shows limited but measurable benefits on structurally similar mazes, highlighting the importance of structural alignment for transfer.

12 Limitations

Key limitations include:

1. Missing BFS targets for extremely dense or disconnected maps.
2. Sensitivity of tabular RL to state-space size and start–goal distribution.
3. Weak generalisation of cluster-specialised Q-tables.
4. Clustering that emphasises obstacle density over higher-order topology.

These limitations inform future directions such as function approximation, goal-conditioned policies, and improved graph-based features.

13 Conclusion

This project provides a complete, reproducible pipeline for unsupervised clustering of navigation environments and adaptive policy switching. We deliver a rich set of features, labels, models, and visual artifacts, with rigorous validation across eight phases. The results confirm that geometry alone can strongly predict map type and difficulty, while RL-based policy transfer remains challenging in high-dimensional grids. The full pipeline and artifacts are archived in the outputs directory for further analysis.

14 Artifacts and Reports

Key outputs are stored in `outputs/figures` and `outputs/reports`. Representative report files include:

- `outputs/reports/classifier_comparison.csv`
- `outputs/reports/classification_map_type_metrics.csv`
- `outputs/reports/classification_difficulty_metrics.csv`
- `outputs/reports/classification_density_category_metrics.csv`
- `outputs/reports/clustering_metrics.csv`
- `outputs/reports/k_selection_metrics.csv`
- `outputs/reports/cluster_profiles.csv`
- `outputs/reports/cluster_vs_maptype.csv`
- `outputs/reports/cluster_vs_difficulty.csv`
- `outputs/reports/cluster_vs_density.csv`
- `outputs/reports/regression_avg_path_length_metrics.csv`
- `outputs/reports/rl_cluster_metrics.csv`
- `outputs/reports/adaptive_vs_single.csv`
- `outputs/reports/reachability_analysis.csv`
- `outputs/reports/master_metrics.csv`
- `outputs/reports/master_metrics_wide.csv`

15 References

- MovingAI Grid Benchmarks: `https://www.movingai.com/benchmarks/grids.html`